

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Third Semester of B. Tech. (CL) Examination****March-April, 2018****CL203 Fluid Mechanics****Date: 04.04.2018, Wednesday****Time: 01:30 p.m. To 4:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the following****[07]**

1. The metacenter is
 - (a) Centroid of displaced fluid volume
 - (b) Mid point between center of gravity and center of buoyancy
2. The study of pressure exerted by water at rest.
 - (a) Hydrostatics
 - (b) Fluid mechanics

(a) A geometrical opening made in side or base of the vessel in such a way that water level is always above the opening is called. (a) Orifice (b) Notch
3. At critical velocity
 - (a) Flow changes from turbulent to laminar
 - (b) Flow changes from laminar to turbulent

(a) A venturimeter is used for measuring (a) Pressure (b) Flow rate
4. The center of buoyancy is
 - (a) Centre of gravity of the body
 - (b) Centre of the displaced fluid volume
5. Mathematical expression for continuity equation of a compressible fluid is given by
 - (a) $AV = \text{constant}$
 - (b) $\rho AV = \text{constant}$

Q - 2(a) Enlist the commonly used methods of dimensional analysis? Discuss the methodology of working with any one of them with suitable example. **[05]**

(b) Attempt any three questions **[09]**

- 1 Enlist fundamental units and demonstrate derivation of any two units from them.
- 2 Enlist forces influencing the motion of a fluid flow. State significance of each force.
- 3 State the mathematical expression of Bernoulli's theorem. Explain each term.
- 4 Water is flowing through a pipe having diameter 300 mm and 200 mm at the bottom and upper end respectively. The intensity of pressure at the bottom end is 24.525 N/cm^2

and the pressure at the upper end is 9.81 N/cm^2 . Determine the difference in datum head if the rate of flow through pipe is 40 lit/s.

- 5 A 25 cm diameter pipe carried oil of specific gravity 0.8 at the rate 120 lit/sec. The pressure at point 'P' 3.5 m above datum line is 19.62 kN/m^2 (gauge) calculate the total energy at point 'P' in meters of oil.

Q - 3(a) A 45° bend is connected in a pipe line, the diameter at the inlet and outlet of the bend is 400 and 200 mm respectively. If the rate of flow is $0.4 \text{ m}^3/\text{s}$ and pressure intensity at inlet of bend is 100 kPa, find the force exerted by water to the bend. **[05]**

(b) Attempt any three questions **[09]**

- 1 A stream function in two dimensional flow is $\psi = 4xy$. Show that the flow is irrotational and determine the corresponding velocity potential ϕ .
- 2 The diameters of a pipe at the sections 1 and 2 are 10 cm and 20 cm respectively. Find the discharge through the pipe if the velocity of water flowing through the pipe section 2 is 10m/s. Also determine the velocity at section 1.
- 3 Obtain the expression of continuity equation for a three dimensional incompressible flow.
- 4 An oil of 5m^3 weights 4000 kg(f). Calculate the specific weight and Specific gravity of this oil.
- 5 A three dimensional flow field described by $V = (x^2+z^2) i + (x^2+z^2) j + (x^2+y^2) k$. Compute the components of acceleration at point (1, 2, 3)

SECTION – II

Q - 4 Answer the following **[07]**

- 1 Define : Match Number
- 2 What is an Isothermal Process?
- 3 Define : Uniform flow
- 4 Name the phenomenon attributable to spherical shape of a rain drop.
 - (a) surface tension (b) Viscosity (c) Specific Gravity (d) None of these
- 5 An ideal fluid
 - a) readily incompressible b) processes viscosity
 - c) is absolutely frictionless and incompressible d) obeys gas laws
- 6 Viscosity of fluid
 - a) Decreases with increase in temperature b) Remains same at all temperature
 - c) Depends upon pressure of the fluid d) Increases with increase in temperature

7 Find out discharge through the pipe of diameter 30 cm with velocity 2.5 m/s.

Q – 5(a) (i) Define pressure head of a liquid and explain with sketch how pressure can be measured using differential U-Tube Manometer. [04]

(ii) Oil of viscosity 4.25 poise flows in a 45 mm diameter pipe. discharge rate is 5.25 lit/s. if specific gravity of oil is 0.87, state whether the flow is laminar or turbulent. [03]

OR

(a) (i) A venturimeter has an area ratio of 9 to 1, the larger diameter being 30 cm. During the flow, the recorded pressure head in the larger section is 6.5 meters and that at the throat 4.25 meters. If coefficient of venturimeter is $C = 0.99$ compute the discharge through it. [04]

(ii) A horizontal pipe 125 mm diameter is joined by sudden enlargement to a 200 mm diameter pipe. Water is flowing through it at the rate $0.075 \text{ m}^3/\text{s}$. Find pressure difference in two pipes in m of water. [03]

(b) (i) Explain how experimentally we can determine the metacentric height. [04]

(ii) A simple manometer is used to measure the pressure of oil of specific gravity 0.8 flowing in a pipe line. Its right limb is open to the atmosphere and the left limb is connected to the pipe. The center of pipe is 9 cm below the level of mercury (sp. gravity 13.6) in the right limb. If the difference of mercury level in the two limb is 15 cm, determine the absolute pressure of the oil in the pipe in kg/cm^2 . [03]

Q - 6 (a) (i) Derive the expression for viscous flow between two parallel plates. [04]

(ii) A submerged sluice is closed by gate having 1.5 m width and 1.2 m height. Difference of water levels on two sides of the gate is 5 m. Find net pressure on gate. [03]

OR

(a) (i) A rocket travels in air of pressure $1.033 \text{ kg}/\text{cm}^2$ at 15°C at velocity of 1650 km/hr. find the Mach number and Mach angle. Take $k = 1.4$ and $R = 29.27$. [04]

(ii) What is Water Hammer effect? Explain occurrence of Water Hammer in a pipe. [03]

(b) (i) How notches are classified based on shape ? Draw a neat sketch of them . With the help of a sketch describe: Nappe and Crest for the notch. [05]

(ii) During an experiment in a laboratory, 0.06 m^3 of water flowing over a right angled triangular notch was collected in one minute. If the head of sill is 54 mm calculate the coefficient of discharge of the notch. [02]
